



Faculty of Engineering and Technology

Electrical and Computer Engineering Department

(ENCS5131)

Project - Configurable Serial-to-Parallel Receiver Using Verilog HDL

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Project Target

Project aims to design a verification environment for a configurable serial-to-parallel receiver using a complete UVM testbench. A receiver is designed to detect start or stop bits. It runs at a fixed baud rate of 1200 bps with a reference clock of 100 MHz. It also optionally checks even parity. The receiver also detects framing or parity errors. It also indicates when data is valid. The receiver also handles data words of 4, 5, 6 or 7 bits long.

Bugs Caught and Fixed in the RTL

#	Bug	Effect	Fix
1	Wrong baud divisor default: CLK _S PER BIT = 104167.	About 960 bps instead of required 1200 bps.	Changed default to 83333.
2	Wrong data size mapping in FSM.	Packets not completed at correct time.	Fixed mapping to 00=4, 01=5, 10=6, 11=7.
3	Shift register reset or clear value was 8'hFF.	Unused bits could contain ones after reset or clear.	Changed reset or clear to 8'h00.
4	LSB first shifting was implemented incorrectly.	Received data bits were reversed or misaligned.	Stored first received data bit in bit 0, then bit 1, etc.
5	Unused upper bits were filled with ones.	Violates zero fill requirement.	Added zero extension for all data sizes.
6	Parity checker used wrong bit ranges.	Parity calculated over wrong number of data bits.	Fixed ranges to 4/5/6/7 active bits.
7	Parity error equation was inverted.	Good parity could fail and bad parity could pass.	Changed error to XOR(data_bits) ^ received_parity.
8	DONE state could hold/stick data_valid.	Back to back packets could fail.	Made data_valid a pulse and returned to IDLE.
9	Parallel data was not cleared on reset.	Reset output did not match cleared behavior.	Reset parallel data to 8'h00.
10	Framing error path could leave stale parity status.	Framing error could show stale parity data.	Cleared parity status on framing error.

UVM Testbench Structure

- **receiver_seq_item**: transaction fields for operation, data size, data, parity control, and expected result.
- **receiver_sequences**: directed tests for valid frames, error frames, and control/corner cases.
- **receiver_driver**: drives serial frames according to the spec: start bit, LSB-first data bits, optional even parity, and stop bit.
- **receiver_monitor**: observes rising edges of data_valid and framing_error.
- **receiver_scoreboard**: compares actual DUT outputs against spec-based expected results.
- **receiver_if** and **receiver_assertions**: interface and SVA checks for reset, valid/error exclusivity, and valid pulse behavior.

Test Cases Summary

Test group	What was tested	Expected result
Reset test	Assert reset while outputs may be unknown or active.	FSM or output status or data clear to safe values.
Valid frame without parity	4, 5, 6, and 7 bit frames with parityenable=0.	Correct right-aligned parallel_data, upper bits zero, data_valid=1, no errors.
Valid frame with even parity	4, 5, 6, and 7 bit frames with correct even parity.	Correct data, data_valid=1, parity_error=0, framing_error=0.
Bad parity frames	4, 5, 6, and 7 bit frames with inverted parity.	parity_error=1 detected.
Framing error	Valid data or parity but stop bit forced to 0.	framing_error=1 and no valid packet accepted.
False start	Serial line low for less than half a bit.	No packet accepted and no output event.
Enable-low frame	Complete frame driven while enable=0.	Packet ignored.
Reset mid-frame	Reset after start bit and some data bits.	State clears and stale packet is blocked.
Back-to-back packets	Two frames sent close together.	Receiver returns to IDLE and accepts both packets.

Expected Final Simulation Result

```
** Report counts by severity
UVM_INFO : 45
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[BAUD_CHECK] 1
[RESET_CHECK] 1
[RESET_MID] 1
[RNTST] 1
[SB_EXP] 18
[SB_PASS] 19
[SB_SUMMARY] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1
[UVM_PASS] 1

$finish called from file "/apps/vcsmx/vcs/X-2025.06-SP1/etc/uvm-1.2/src/base/uvm_root.svh", line 532.
$finish at simulation time 121041825000
VCS Simulation Report
```